
CLAM: **Causal Spatial Disaggregation** **to Infer Local Effects From Coarse Data**

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Abstract

1 Learning fine-grained spatial patterns from coarse-resolution data is challenging,
 2 especially in causal settings where high-resolution effects must be inferred from
 3 aggregated interventions and outcomes. We introduce CLAM, a method for estimat-
 4 ing localized causal effects from coarse observations by exploiting high-resolution
 5 contextual covariates that modulate these effects. By jointly learning the causal
 6 mechanism and a disaggregation mapping, CLAM captures interactions that are
 7 missed when addressing these problems independently. The method supports
 8 localized effect estimation, counterfactual reasoning, and principled outcome dis-
 9 aggregation, and reliably captures spatially varying causal effects across diverse
 10 settings. This is particularly relevant for applications such as public health and
 11 environmental policy, where decisions are made at broad scales despite substantial
 12 local heterogeneity. Code is available at github.com/DSanonym/clam.

1 Introduction

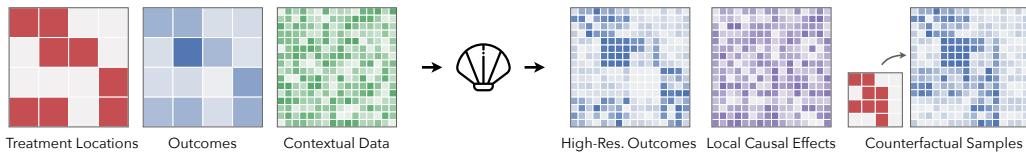


Figure 1: **Motivating example.** CLAM takes regional treatments (e.g., a vaccination awareness campaign) and outcomes (e.g., vaccination levels), along with subregional covariates (e.g., demographics), to learn treatment–outcome relationships and estimate local causal effects or sample counterfactuals.

14 Going from high-resolution (HR) to low-resolution (LR) data is typically straightforward, as it can
 15 be done via aggregation or pooling functions such as sums or means. These operations generally
 16 remove information. In contrast, going from LR to HR data is fundamentally difficult because it
 17 requires restoring information that has been lost. This generally necessitates some form of prior
 18 knowledge about the HR data. This inverse problem is typically studied under terms such as *spatial*
 19 *disaggregation* [1], *downscaling* [2], *ecological inference* [3], or *super-resolution* [4]. It is relevant
 20 across many fields involving spatio-temporal statistics, including earth science [5], public health [6],
 21 and the social sciences [7].

22 Causality [8], on the other hand, deals with the effect of interventions on outcome variables. Its core
 23 insight is that statistical associations do not imply causal relationships. In other words, observing a
 24 system is not the same as knowing how intervening on it will change its dynamics.

25 Answering causal questions is often attempted using LR data, but this data is frequently insufficient
 26 for this [9]. For example, interventions, like vaccination campaigns, might be applied across broad
 27 regions, yet their impact could vary significantly at the subregional level. Surprisingly little effort has
 28 been made to properly combine spatial disaggregation with the quest to answer causal questions.

29 **Contribution.** This work addresses the gap between spatial disaggregation and causal inference
 30 by introducing a unified framework grounded in structural causal models (SCMs) and Pearl’s do-
 31 calculus [8]. We argue that jointly tackling causal inference and disaggregation is natural in many
 32 real-world settings, where interventions are implemented at coarse scales but effects manifest locally.
 33 While our primary focus is the formulation of this framework, we illustrate its capabilities and
 34 limitations through a series of simulation studies that mimic realistic data-generating processes, as
 35 well as a case study based on real-world data on the impact of heat waves on gun violence.

36 CLAM (Causal disaggregation method) can be summarized as follows: We learn a mechanistic func-
 37 tion that maps an intervention indicator and contextual information to high-resolution (HR) outcomes.
 38 In doing so, we treat HR contextual data (or *covariates*) as an auxiliary variable that implicitly defines
 39 a prior over the HR outcome (cf. Figure 1). The model is trained by aggregating the predicted HR
 40 outcomes and enforcing consistency with the observed low-resolution (LR) measurements. This
 41 enables causal effect estimation, counterfactual reasoning, and the computation of HR estimates from
 42 LR inputs.

43 2 Problem Setup

44 For notational simplicity and without loss of generality, we assume that all regions share the same
 45 subregion structure, interventions are binary, and all variables are scalar rather than vector-valued.

46 We consider N regions, each divided into M subregions. The LR *treatment data* is a vector
 47 $\hat{T} \in \{0, 1\}^N$, where each entry $\hat{t}_i$ indicates whether an intervention occurred in region i . We assume
 48 that, when present, the intervention affects all subregions within that region.

49 The LR *outcome data* is a vector $\hat{Y} \in \mathbb{R}^N$, obtained by applying a row-wise aggregation function
 50 (e.g., summation or averaging) to an unobserved HR outcome matrix $Y \in \mathbb{R}^{N \times M}$, where $y_{i,j}$ denotes
 51 the outcome value for subregion j within region i . The HR *contextual data* (covariates) is given by a
 52 matrix $C \in \mathbb{R}^{N \times M}$, where $c_{i,j}$ denotes the contextual value for subregion j in region i .

53 Finally, we assume that N and M are perfect squares, allowing us to represent the spatial structure as
 54 a $\sqrt{N} \times \sqrt{N}$ grid of regions, each containing a $\sqrt{M} \times \sqrt{M}$ grid of subregions (see Figure 1 for an
 55 example with $N = M = 16$ and $\sqrt{N} = \sqrt{M} = 4$).

56 **Goal.** Our goal is to learn a function that estimates local causal effects and to disaggregate coarse
 57 outcomes into fine-grained predictions. We formalize this objective in the next section.

58 **Generalizations.** The setup admits several natural extensions. Variables (such as $c_{i,j}$) can be
 59 vector-valued, treatments can be continuous instead of binary, and the observed LR treatment $\hat{T}$
 60 can be interpreted as an aggregation of an underlying HR treatment matrix that varies within a
 61 region. The framework also extends to spatio-temporal settings, where T , Y , and C evolve over time.
 62 More generally, additional structure such as hidden confounding, mediation, or non-trivial treatment
 63 assignment can be incorporated; we discuss these cases in Section 4. Naturally, the spatial layout
 64 need not follow a regular grid, which we adopt only for illustration.

65 **Causal Assumptions.** Depending on the experimental setting, different assumptions may hold
 66 or be relaxed. We may assume a known causal graph, which constrains the model and reduces the
 67 risk of spurious explanations, particularly important when contextual variables are high-dimensional
 68 and correlated, but only a subset is causally relevant. We typically assume no hidden confounding,
 69 meaning all relevant variables affecting the outcome are observed, allowing interventions to be
 70 interpreted causally up to reasonable noise. Causal effect identification also requires variation in
 71 interventions across regions. If the aggregation function from subregional to regional outcomes is
 72 known (e.g., a mean or sum), disaggregation becomes more tractable. Moreover, we typically assume
 73 that the treatment does not depend on the context values.

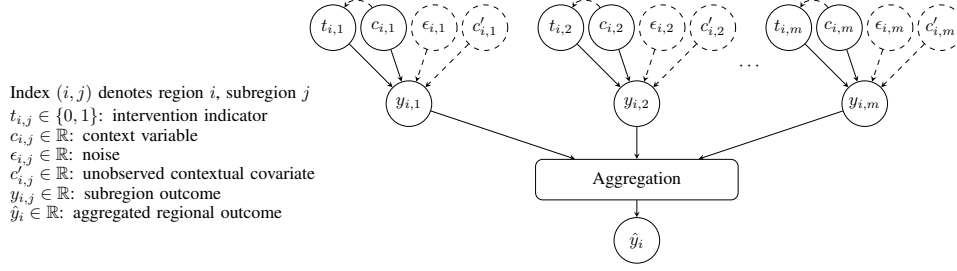


Figure 2: Causal graph for one region. All functional relations are shared among subregions and regions. Variables could also be vector-valued. Unobserved covariates and a causal link from context to treatment are optional. The aggregation is typically a mean or a sum.

3 Our Method: CLAM

Our method combines two insights. First, we represent the causal disaggregation problem using a structural causal model formalism. This formalism encodes assumptions and specifies the missing pieces (functions, parameters). Second, we train an ML model to infer the missing pieces where we use the LR data as a loss signal to estimate the most likely HR data and functional relationships.

3.1 Structural Causal Model

Each region–subregion pair (i, j) is associated with a treatment $t_{i,j}$, context $c_{i,j}$, noise $\epsilon_{i,j}$, and an optional hidden confounder (or non-confounding covariate) $c'_{i,j}$ (cf. Figure 2). A shared structural function maps these inputs to a subregional outcome $y_{i,j}$, and then aggregates the subregional outcomes to a regional outcome $\hat{y}_i$:

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}, \epsilon_{i,j}, c'_{i,j}), \quad \hat{y}_i = g_{\phi}(\{y_{i,j}\}_{j=1}^M).$$

Each subregion is modeled by a local functional mechanism, with aleatoric uncertainty captured via $\epsilon_{i,j}$. Although the functions $f_{\theta}(\cdot)$ and $g_{\phi}(\cdot)$ are shared across regions, region-specific dynamics can still be modeled through context variables (e.g., region indicators or positional encodings).

In general, $g_{\phi}(\cdot)$ can be any function parameterized by ϕ . For simplicity, we typically assume the aggregation is a sum, the noise is additive, and there are no hidden confounders. Then the model simplifies to:

$$\hat{y}_i = \sum_{j=1}^M (f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j}) = \sum_{j=1}^M f_{\theta}(t_{i,j}, c_{i,j}) + \sum_{j=1}^M \epsilon_{i,j}.$$

3.2 Training

The model is trained by minimizing the discrepancy between the observed LR outcomes $\hat{Y}$ and the predicted aggregated outcomes $\hat{y}$, typically using an MSE loss: $\mathcal{L}_{\theta, \phi} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - \hat{Y}_i)^2$. Note that we use $\hat{Y}_i$ for the observed outcome in region i and $\hat{y}_i$ for the corresponding prediction.

Importantly, the loss is only observed at the aggregated (region) level. The model therefore does not receive direct supervision for individual subregions; instead, it must learn a subregional mechanism that, when combined, is consistent with the observed aggregated outcomes.

When relaxing the assumptions from Section 2, training becomes more complicated. For example, if the aggregation function, $g_{\phi}(\cdot)$, is unknown, it must be learned jointly with the local mechanism $f_{\theta}(\cdot)$, which leads to a coupled estimation problem. In general, $g_{\phi}(\cdot)$ can be implemented using an (order-invariant) set neural network (e.g., [10]). This is conceptually similar to settings in graph neural networks, where both the message-passing mechanism and the aggregation function are learned [11].

Similarly, if treatments are not uniformly applied across subregions, the underlying HR treatment matrix must be inferred jointly during training. More generally, additional latent variables (e.g., unobserved confounders) can be incorporated and estimated within the same framework.

3.3 Downstream Applications

Training yields the predictive mechanism $f_{\theta}(\cdot)$, from which several causal quantities can be derived. The most direct is what we call the *local conditional average treatment effect* (LOCATE), a version of the conditional average treatment effect (CATE) conditioned on subregion-specific factors:

$$E_{i,j} = f_{\theta}(1, c_{i,j}) - f_{\theta}(0, c_{i,j}), \quad (1)$$

i.e., the treatment–control difference at the subregional level. Aggregating these quantities yields ATEs for regions, populations, or environments; with continuous treatments, the same mechanism gives conditional dose–response curves. Note that LOCATE is noise-marginalized.

Beyond effect estimation, the framework supports outcome *disaggregation*—mapping coarse regional outcomes back to fine-grained predictions—and counterfactual reasoning, e.g., predicting outcomes under hypothetical intervention placements or alternative contexts. In the simplest case, this amounts to evaluating $f_{\theta}(\cdot)$ under new inputs. For counterfactuals, when the noise term is believed to carry realization-specific information, inferred noise values can be reused to produce sharper, sample-specific counterfactuals rather than population-level averages.

The framework can also recover latent HR treatment locations by representing them as trainable variables (cf. Exp. 2 in Appendix C.2). With time-series data, CLAM can further infer latent variables that modulate causal effects, provided temporal variation supplies enough constraints (cf. Exp. 3 in Appendix C.3).

3.4 Relationship to Causal Deabstraction

We view CLAM as an instance of *causal deabstraction*: recovering a fine-grained causal mechanism from coarse, aggregated observations. While prior work studies when causal structure is preserved under abstraction [9, 12], we consider the inverse problem: inferring a high-resolution model that is consistent with observed aggregates. Appendix A formalizes this view and clarifies the causal semantics.

The key idea is that aggregation does not fully erase causal information when (i) the aggregation process is shared and (ii) regions differ in their subregional compositions.

Under the assumption of invariant causal mechanisms [13, 14], this variation induces constraints that can make the underlying mechanism estimable in favorable settings, although formal identifiability depends on the aggregation function, function class, covariate diversity, and noise model.

4 Exploiting Context Across Motifs in the Causal Graph

Throughout the paper, we focus on the simplest causal motif: a treatment T acting on an outcome Y , with a contextual covariate C that modulates the effect. Real-world settings can exhibit a much richer variety of causal structures, and the role of high-resolution context, C , shifts with the motif: sometimes it must be conditioned on, sometimes ignored, sometimes leveraged for new identification strategies. Do-calculus tells us *whether* an effect is identifiable given the graph; this section asks *how* the LR/HR distinction interacts with that question. Here, we provide an outlook in which we sketch how HR context can be exploited in common causal graph motifs.

Baseline: C as a Pure Effect Modifier ($T \rightarrow Y, C \rightarrow Y$). The default setting throughout this paper is the simplest motif: T and C are independent inputs, both pointing into Y , with no edge between them. C is not a confounder, only a covariate that modulates the local effect of T , and the backdoor criterion holds trivially and gives rise to the LOCATE in Eq. (1). The role of the HR resolution is purely informational: C provides multiple “views” of the same regional outcome and thereby constrains the shared mechanism $f_{\theta}(\cdot)$.

Confounder: $C \rightarrow T, C \rightarrow Y, T \rightarrow Y$. When the context C also drives whether (or by how much) a region is treated, it becomes a confounder. The backdoor criterion then requires conditioning on C . Treating T as exogenous yields biased estimates; modeling the treatment-assignment mechanism at HR is what makes the adjustment correct. Exp. 5 (Appendix E) instantiates this motif and shows that ignoring the dependence inflates the MSE on local effects. A concrete real-world example to

152 illustrate this problem is a wildfire-prevention program: forested counties with dry summers are more
 153 likely to receive funding, and those same counties also have higher baseline fire incidence, so the
 154 program’s effect is conflated with the underlying risk.

155 **Collider:** $T \rightarrow C, Y \rightarrow C$. A collider is a variable influenced by both the treatment and the
 156 outcome. Conditioning on such a variable opens a spurious path between T and Y , so naïvely passing
 157 C as an input to $f_{\theta}(t, c)$ would bias the estimated effect. Nevertheless, HR collider information
 158 may still be useful: since C is partly determined by the HR data $y_{i,j}$, it can provide fine-grained
 159 information for the reconstruction of Y or serve as a diagnostic check. For example, consider a
 160 regional vaccination campaign (T), where the outcome of interest is infection rates (Y), and the HR
 161 variable C is neighborhood-level pharmacy purchases of rapid tests. The campaign may increase
 162 public attention and test purchases through its messaging, while actual local infections also increase
 163 demand for rapid tests.

164 A possible way to exploit this structure is to invert the usual roles: instead of using C as a covariate of
 165 the outcome, we treat C itself as the HR target and learn a local mechanism $f^{\text{inv}}(t, y) = c$ that maps
 166 treatment and outcome to the collider. Under this setup, CLAM’s aggregation-consistency principle
 167 still applies, but it is used to jointly learn the mechanism and the disaggregation of T and Y against
 168 the observed HR values of C . We leave a concrete implementation to future work.

169 **Mediator:** $T \rightarrow C, C \rightarrow Y, T \rightarrow Y$. A mediator C lies on the causal pathway from T to Y , so
 170 part of the treatment effect is transmitted through C rather than acting directly. Naïve estimation that
 171 conditions on C cannot separate the direct effect of T from the indirect effect through C . When the
 172 high-resolution mediator is observed, one can instead jointly learn the treatment-to-mediator, mediator-
 173 to-outcome, and direct treatment-to-outcome mechanisms from aggregated regional observations.
 174 This enables a decomposition of the learned effect into a direct component $T \rightarrow Y$ and an indirect
 175 component $T \rightarrow C \rightarrow Y$. A concrete example is a regional advertising campaign observed at low
 176 resolution, whose effect on regional sales, also observed at low resolution, is partly direct and partly
 177 mediated by word-of-mouth dynamics reflected in local online discourse observed at high resolution.

178 **Spill-over.** A subregional outcome may depend on neighboring treatments and contexts: interven-
 179 tions can spill over across space. The local mechanism is then no longer purely local, and aggregated
 180 observations alone cannot distinguish a strong local effect from a weaker effect that diffuses outward.
 181 HR context can help by exposing spatial structure; technically, one can replace the pointwise $f_{\theta}(\cdot)$
 182 with a grid-based model. In principle, the model can learn both direct local effects and spill-over
 183 patterns from aggregated outcomes. A concrete example is a driving ban in one district that displaces
 184 traffic into adjacent ones, so regional air quality depends on both the ban location and surrounding
 185 road and vegetation patterns.

186 **Time-Varying Confounding.** In spatiotemporal settings, confounders may evolve over time and
 187 be shaped by past treatments. Present context can then no longer be treated as a fixed pre-treatment
 188 covariate, because it may already encode earlier interventions. This makes estimation harder: naïve
 189 adjustment can block part of the causal pathway or introduce bias [15]. CLAM can be extended
 190 by indexing $f_{\theta}(\cdot)$ over time and by explicitly modeling how past treatments, outcomes, and HR
 191 covariates affect the current state. Exp. 3 already uses temporal variation to recover a stable latent
 192 spatial factor; a natural next experiment would make this factor dynamic and treatment-dependent.
 193 For example, neighborhoods with high prior infection rates may receive additional public-health
 194 resources, which then change later vulnerability and influence future allocation decisions.

195 **Summary.** Across all motifs, the role of HR data can be summarized as follows: it provides
 196 *variation within regions* that translates into observable differences at the aggregated level. This
 197 variation can be exploited to constrain and identify fine-grained causal mechanisms, but only when
 198 the causal graph permits it.

199 More concretely: (i) for effect modifiers and confounders, HR structure is highly informative and
 200 can be directly leveraged for identification and adjustment; (ii) for mediators, it remains informative
 201 but requires explicitly modeling the underlying causal pathways; (iii) for colliders, it can be actively
 202 misleading if the framework is not changed accordingly.

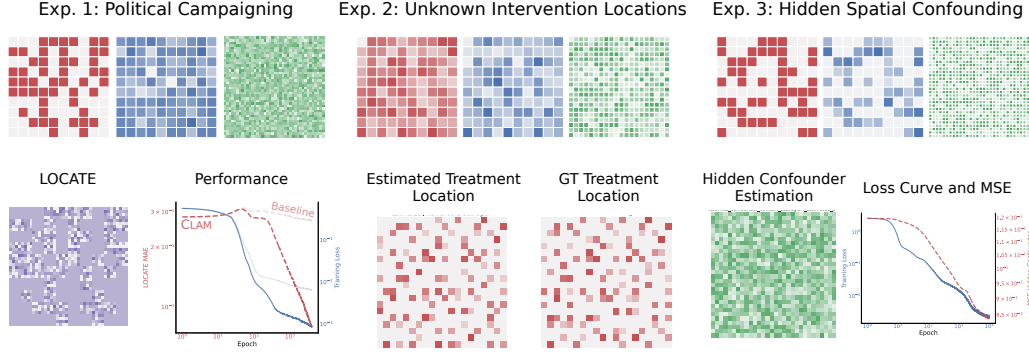


Figure 3: **Results Overview.** **Top:** Input of LR intervention locations (red), LR outcomes (blue), and HR context (green). **Bottom:** Selected inference results.

CLAM does not replace causal identification via do-calculus: the causal graph determines whether a target effect is identifiable in principle, while CLAM provides a mechanism for estimating the corresponding fine-grained structure from aggregated data.

However, identifiability in principle does not guarantee successful recovery in practice. Learning the underlying mechanism further requires sufficient variability in the HR covariates, a well-behaved optimization problem, and effective training of the chosen function class. We discuss these limitations in more detail in Section 7.

5 Related Work

CLAM sits at the intersection of multiple research strands; an extended discussion is provided in Appendix B.

Spatial Disaggregation and Downscaling. A long line of work in statistical downscaling and spatial disaggregation infers high-resolution estimates from coarse data using interpolation, regression, or learning-based techniques [1, 2, 16, 17]. These methods are predominantly associational; recent extensions integrate causally relevant predictors [18, 19] but still rely on relatively fine-grained information and do not target interventional or counterfactual queries.

Causal Inference under Aggregation and Abstraction. Reasoning causally with aggregated data is well known to be hazardous: classical results on the ecological fallacy show that aggregate statistics may obscure or invert relationships at the unit level [20–23]. Work on causal abstraction formalizes when mappings between models at different granularities preserve interventional or counterfactual semantics [9, 24, 12], and spatiotemporal causal inference [25–31] addresses related challenges with different data regimes. None of these directly target the joint problem of disaggregating outcomes and estimating local causal effects from coarse interventions.

Invariant Mechanisms, GNNs, and Hierarchical Models. Conceptually, CLAM relies on the principle of *invariant causal mechanisms* [13, 14, 32, 33]: the local mechanism $f_{\theta}(\cdot)$ is shared across subregions while contextual covariates drive heterogeneity. This turns aggregation into a constraint and connects our architecture to message-passing GNNs [11] and Deep Sets [10], as well as to hierarchical Bayesian models that pool information across groups [34, 35].

6 Experiments

We evaluate CLAM on three synthetic studies and one real-world study. In the synthetic studies the ground truth mechanism is available isolate distinct capabilities of the framework under controlled conditions: heterogeneous local-effect recovery, latent intervention localization, and the reconstruction of unobserved covariates. The real-world study then probes the method on U.S. gun-violence data, where we assume that only coarse regional outcomes are observed.

Full experimental details, per-experiment figures, and additional analyses are provided in Appendices C and D, with the ablation study included in Appendix C. Two additional studies—covering confounded treatment allocation and an unknown aggregation function—are deferred to Appendix E. Our focus is qualitative: we aim to characterize the behavior of the method. Figure 3 provides an overview of the inputs and headline results. Code that runs off-the-shelf on Google Colab is provided.

6.1 Synthetic Studies

Exp. 1: Political Campaigning. We first ask whether CLAM can recover heterogeneous local causal effects from aggregated outcomes when the effect of treatment depends on subregional context. We model a binary regional intervention (campaign spending) whose impact varies with subregional wealth (z-scored income values). Regional-level covariates largely remain uninformative by construction. The reported outcome is the candidate’s relative improvement, observed only at the regional level. We compare against a naïve **baseline** (Figure 3, with confidence intervals in the Appendix) that applies a uniform disaggregation mechanism to the regional outcome and then learns the causal mechanism. We use a simple neural network to estimate the causal mechanism. CLAM accurately recovers the underlying subregional causal effects, with the MAE to the ground-truth LOCATE matrix (cf. Eq. (1)) decreasing alongside the training loss. The learned model supports coherent counterfactual reasoning at both regional and subregional resolutions and correctly identifies regions whose outcome differences arise purely from contextual heterogeneity rather than mean effects. Full setup and per-subregion visualizations are given in Appendix C.1 (Figure 5).

Exp. 2: Public School Funding. Next, we test whether CLAM can recover *latent intervention locations* jointly with local causal effects when only aggregated intervention totals and outcomes are observed. Each of the 30×30 regions consists of 4×4 subregions and receives binary funding in either no subregion or exactly one unobserved subregion. Treatment effects vary with subregional socioeconomic status, and the outcome is education-score improvement. We implement $f_\theta(\cdot)$ using either a parametric function or a small neural network. In both cases, CLAM jointly identifies the treated subregions and accurately recovers the underlying causal-effect function. To sample a treated subregion, we use the common Gumbel-Max trick. Details are provided in Appendix C.2 and Figure 7.

Exp. 3: Spatiotemporal Effects of Heat Waves. Finally, we ask whether CLAM can reconstruct an *unobserved spatial effect modifier* from aggregated outcomes in a spatiotemporal setting. Heat waves negatively affect school performance, with heterogeneous impact depending on parental education (observed) and vegetation coverage (unobserved), the latter mitigating heat exposure. Only region-level outcomes are observed over time. Under a correctly specified linear mechanism, both the regional prediction loss and the MSE converge rapidly, and the estimated disaggregation of the vegetation is visually indistinguishable from the ground-truth vegetation map. With a flexible MLP, CLAM still recovers meaningful spatial structure but with substantially higher error. This reflects an identifiability trade-off whereby expressive models can absorb transformations of the vegetation while preserving aggregated predictions. Details are provided in Appendix C.3 (Figure 8).

Takeaways. Across the three studies, CLAM performs well in the synthetic setting, particularly for estimating causal effects. The recovery of latent variables, such as intervention locations or hidden confounders, depends on the specific experiment. Overfitting may occur, when large amounts of data are available.

6.2 Semi-Synthetic Case Study With Real-World Data: Heat and Gun Violence

We evaluate CLAM on U.S. gun-violence data from the Gun Violence Archive (GVA) for 2022–2023 [36], motivated by prior evidence linking higher ambient temperatures to increased firearm violence [37]. We treat temperature as a continuous exposure. Urbanization is the high-resolution (HR) contextual covariate, and gun-incident counts are observed at weekly resolution. A core difficulty in evaluating disaggregation methods on real data is the absence of ground truth at the fine scale. We work around this by starting from data that is natively HR (incident counts at the cell level) and *artificially* aggregating it into LR regional counts, which serve as the inputs to CLAM. To construct a high-quality reference **baseline**, the original HR counts are only used for fitting an “oracle” that we use for evaluation.

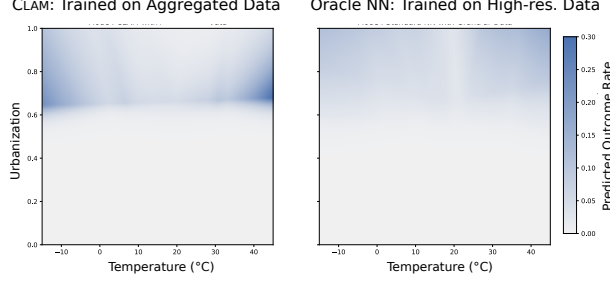


Figure 4: Side-by-side comparison of the learned rate functions $f_{\theta}(\text{temp}, \text{urban})$ for the disaggregation from CLAM (left) and the oracle (a NN with the ground truth subregional data; right). Predictions are generated across a temperature range of $[-15, 45]^{\circ}\text{C}$ and an urbanization scale of $[0, 1]$.

We use $N = 100$ regions and $M = 100$ cells per region, indexed by region i , subregion j , and week w . Let $Y_{i,w}^{\text{LR}}$ be the aggregated regional count in region i during week $w \leq 52$, $c_{i,j}$ the HR urbanization covariate, and $t_{i,w}$ the standardized regional temperature. We model subregion-level rates as $\lambda_{i,j,w} = f_{\theta}(\tilde{t}_{i,w}, c_{i,j})$ and aggregate them as $\mu_{i,w} = \sum_{j=1}^M \lambda_{i,j,w}$, with $Y_{i,w}^{\text{LR}} \sim \text{Poisson}(\mu_{i,w})$. Training minimizes the regional Poisson negative log-likelihood. Figure 4 shows that CLAM recovers a qualitatively similar temperature–urbanization interaction to the oracle. The spatial comparison in Appendix Figure 9 shows the same pattern: plausible structure, but inflated and more diffuse local intensities, reflecting an aggregation-induced identifiability gap.

7 Limitations

CLAM relies on assumptions that are necessary given the nature of inferring fine-grained causal effects from aggregated data, but are nonetheless restrictive. We discuss the main limitations here.

Prerequisite: Do-Calculus Identifiability. CLAM is not a substitute for causal identification. The method should only be applied in settings where do-calculus (applied to a known or expert-specified causal graph) already certifies that the causal mechanism is in principle recoverable. If the causal graph is unknown or the required identification conditions fail (e.g., the backdoor criterion is violated by unmodeled confounders), CLAM will fit a model to the data that will not carry a causal interpretation.

No Identification Guarantee from Optimization Alone. Even when do-calculus certifies identifiability in principle, we do not provide a formal guarantee that our optimization procedure recovers the true causal effect. The training objective is non-convex, and the best we can reasonably claim is that, under good initialization and sufficient covariate diversity, gradient descent finds a local minimum whose value is close to the true causal effect. It is easy to come up with pathological cases in which different fine-grained disaggregations are equally consistent with the observed regional outcomes. How to detect or avoid them in practice, remains an open question.

Assumptions and Their Scope. Recovery of the local mechanism further requires (i) an invariant causal mechanism across subregions, (ii) sufficient diversity in subregional covariate distributions across regions, (iii) no hidden confounding, or its explicit modeling, and (iv) a known or learnable aggregation function.

The invariance assumption (i) is less restrictive than it may appear: although $f_{\theta}(\cdot)$ is shared across regions and subregions, region-specific dynamics can still be captured through contextual covariates such as region indicators or positional encodings. Apparent heterogeneity across regions is absorbed by the context, yielding a *conditionally* invariant model. The stronger requirement is that the contextual covariates sufficiently capture all sources of heterogeneity that are correlated with the treatment. Unobserved factors that are uncorrelated with the treatment effectively act as additional noise: they make training harder but do not bias the estimated effect. Unobserved confounders, by contrast, will generally lead to incorrect estimates unless they are explicitly modeled. The aggregation function (iv) need not be perfectly known in advance: as demonstrated in Exp. 4 (Appendix E), it can

be jointly learned alongside the local mechanism provided one knows that the aggregation itself is uncertain and parameterizes it accordingly.

Identifiability Gap and Latent Variable Recovery. Even when all assumptions hold, aggregation induces an intrinsic identifiability gap: the constraint that aggregated predictions match aggregated observations may admit multiple equally valid solutions. Our real-world study illustrates this to some extent, with CLAM systematically inflating local effect magnitudes relative to the cell-supervised oracle. The problem is amplified when latent variables are jointly inferred (e.g., intervention locations in Exp. 2, unobserved covariates in Exp. 3): the training landscape becomes less smooth, making it harder to find a good optimum, and different inferred values of the latent variables can each correspond to a mechanistic function that explains the data equally well. A rigorous theoretical treatment of identifiability, together with practical diagnostics for detecting violations, is the most important direction for future work.

Limited Real-World Validation. Our evaluation is primarily qualitative and relies substantially on synthetic studies, where the data-generating assumptions are known and can be matched to the model. The real-world study uses a subregion-supervised oracle as a reference rather than a true causal benchmark. However, the causal interpretation of temperature on gun-violence intensity is supported by prior literature [37]. Robustness to realistic noise, measurement error, nonstationarity, and treatment–context dependence is explored in the additional experiments, but a more systematic benchmarking remains an important direction for future work.

8 Conclusions and Future Work

The primary contribution of this work is not a single algorithm but the articulation of a *problem class*: estimating high-resolution causal effects from low-resolution interventions and outcomes by exploiting high-resolution context. We frame this as *causal deabstraction*, inverting the standard notion of causal abstraction and recasting spatial aggregation as a constraint satisfaction problem, thereby bridging structural causal models and spatial statistics. In many real-world settings such as public health, environmental policy, and education, spatial disaggregation and causal estimation are inseparable, since interventions are implemented at coarse scales but their effects manifest locally. Treating them jointly within a structural causal model makes assumptions explicit, clarifies what is and is not identifiable, and turns causal inference into a tractable optimization problem.

CLAM is one concrete instantiation of this framework. It explicitly adjusts for subregional treatment–context confounding using only coarse regional supervision, a historically challenging problem in ecological inference, and jointly learns latent high-resolution variables such as exact intervention locations and spatial confounding alongside the causal mechanism. Across synthetic and real-world studies, we recover meaningful spatial and functional structure from aggregated data, while exposing the inherent limits of the setting: relative structure and causal dependencies are recoverable, but absolute local magnitudes may be systematically overestimated without fine-grained outcome supervision. We view this not as a flaw of CLAM but as a property of the problem class itself, and the identifiability gap it implies is an invitation rather than a verdict. Because the problem class is, to our knowledge, new in this explicit form, principled baselines are scarce, and we hope this paper motivates methods that supersede ours.

Although this paper primarily proposes a research framework, misuse in high-stakes domains such as epidemiology could cause harm if poorly identified local effects guided interventions or resource allocation; outputs should be accompanied by uncertainty estimates, sensitivity analyses, and domain expertise. Disaggregation itself may also raise privacy concerns, since inferred high-resolution patterns can reveal sensitive local information.

Several directions stand out. Theoretically, the most pressing need is a rigorous identifiability analysis: which combinations of aggregation function, contextual diversity, and structural restrictions guarantee recovery of the local mechanism, and which only up to monotone or scale transformations? Methodologically, principled regularization, hierarchical priors, or Bayesian formulations could control aggregation-induced bias and provide uncertainty quantification. Relaxing the shared-mechanism assumption, integrating causal discovery over contextual variables, incorporating instrumental-variable settings, and extending to spatiotemporal aggregation are natural next steps. We hope that framing CLAM as a first concrete entry in a broader problem class makes these extensions easier to pursue.

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473 A Causal Deabstraction Framework

474 In this section, we situate our work within the broader causal inference literature. Specifically, we
 475 frame this as a novel instance of *causal deabstraction* [9, 12]. Our methodology is informed by the
 476 framework of SCMs and is guided by the principle of ICM.

477 A.1 Structural Causal Framework

478 The structural causal model introduced in the previous section 3.1) can be formalized in generative
 479 form as follows. Each spatial unit (region) $i \in \{1, \dots, N\}$ is composed of M subregions indexed by
 480 $j \in \{1, \dots, M\}$. For each subregion (i, j) , we posit the following SCM:

$$\begin{aligned} c_{i,j} &\sim P_C \\ \epsilon_{i,j} &\sim P_\epsilon \\ t_{i,j} &= T_i \in \{0, 1\} \\ y_{i,j} &= f_\theta(t_{i,j}, c_{i,j}, \epsilon_{i,j}) \\ \hat{y}_i &= g_\phi(\{y_{i,j}\}_{j=1}^M) \end{aligned}$$

481 Here, $c_{i,j}$ denotes a high-resolution contextual covariate for subregion j within region i , drawn
 482 from distribution P_C , and $\epsilon_{i,j}$ represents exogenous noise drawn from P_ϵ . The binary intervention
 483 $t_{i,j} = T_i \in \{0, 1\}$ is assigned at the regional level and shared across all subregions. The function
 484 $f_\theta(\cdot)$ encodes the local causal mechanism—shared across all subregions—that maps the intervention,
 485 context, and noise to the subregional outcome $y_{i,j}$. The aggregation function $g_\phi(\cdot)$ maps the collection
 486 of subregional outcomes $\{y_{i,j}\}_{j=1}^M$ to the regional outcome $\hat{y}_i$.

487 In the main section, we considered simplified cases—such as sum aggregation, additive noise, and
 488 the absence of hidden confounders—to provide intuition. The formulation here generalizes that view
 489 by expressing the structural causal model in explicit stochastic form, specifying how interventions,
 490 contextual variables, and noise are generated, while making clear that both $f_\theta(\cdot)$ and $g_\phi(\cdot)$ are shared
 491 across regions.

492 A.2 Invariant Causal Mechanism Assumption

493 A central assumption in CLAM is that the causal function $f_\theta(\cdot)$ is *invariant* across all subregions and
 494 regions. That is, while the distribution of covariates $\{c_{i,j}\}$ may vary across regions due to differences
 495 in local composition, the functional form $f_\theta(t, c)$ does not change. This assumption reflects the
 496 principle of *invariant causal mechanisms* [14, 13, 32], which posits that causal relations are stable
 497 across environments unless directly intervened upon.

498 We treat each region i as an *environment* defined by its distribution over covariates $\{c_{i,j}\}$, but
 499 governed by a shared mechanism $f_\theta(\cdot)$. The variability in covariate composition across regions
 500 allows us to disentangle $f_\theta(\cdot)$ from the observed aggregated outcomes y_i .

501 A.3 Deabstraction via Invariance

502 We propose to view the task of estimating $f_\theta(\cdot)$ from aggregated outcomes as an instance of *causal*
 503 *deabstraction* that is, inferring a latent, high-resolution causal model that is consistent with a lower-
 504 resolution aggregated model. This perspective is dual to recent work on causal abstraction [9, 12],
 505 which studies when a coarse model preserves the counterfactual semantics of a fine-grained one.
 506 Here, we reverse the direction: we seek to recover a fine-grained causal mechanism that is *compatible*
 507 with the observed coarse-level effects.

508 This is possible due to two key properties:

- 509 1. The aggregation function $g_\phi(\cdot)$ is consistent and applied uniformly across regions.
- 510 2. The diversity in the covariate compositions $\{c_{i,j}\}_{j=1}^M$ across regions induces identifiable
 511 variation in the aggregates y_i under the shared mechanism $f_\theta(\cdot)$.

512 Given sufficient diversity and the assumption of causal invariance, the aggregated outputs provide a
 513 supervisory signal to recover the latent causal function.

514 A.4 Causal Semantics via Do-Calculus

515 Under the assumption of no hidden confounding, the causal effect of the intervention T_i on a
516 subregional outcome $y_{i,j}$ is identified via the backdoor criterion:

$$\mathbb{E}[y_{i,j} \mid do(T_i = t)] = \int \int f_{\theta}(t, \epsilon_{ij}) p(c_{ij}) p(\epsilon_{ij}) dc_{ij} d\epsilon_{ij}$$

517 In Exps 1–4, we assume $T_i \perp\!\!\!\perp \epsilon_{i,j} \mid c_{i,j}$, and that T_i is either randomized or independent of $C_{i,j}$.
518 Under this setting, the expectation can be approximated via observational data. However, because
519 only the aggregate outcome y_i is observed, we must rely on the consistency constraint:

$$y_i \approx \sum_{j=1}^M f_{\theta}(T_i, c_{i,j})$$

520 The training procedure thus finds the function $f_{\theta}(\cdot)$ that jointly explains all observed aggregates under
521 this constraint, while enforcing that $f_{\theta}(\cdot)$ is invariant across regions. In this sense, the estimation of
522 $f_{\theta}(\cdot)$ becomes a constraint satisfaction problem, where causal invariance and aggregation consistency
523 define the feasible solution space. In Exp 5, this conditional independence assumption is deliberately
524 violated by construction, and valid estimation requires explicitly modeling the treatment allocation
525 mechanism.

526 A.5 Implications

527 The assumption of an invariant subregional mechanism plays a central role in enabling causal
528 inference in our disaggregated setting. By treating the coarse observations as aggregates of fine-
529 grained causal effects and leveraging variation in covariate compositions, CLAM performs inference
530 not merely on statistical associations but on causal mechanisms that are meaningful and robust to
531 changes in population structure. All in all, we contribute a formal and algorithmic instantiation of
532 *causal deabstraction* from low-resolution outcomes and high-resolution covariates.

533 B Extended Related Work

534 Our work builds on and extends three main strands of research: spatial disaggregation, causal
535 inference under aggregation, and causal abstraction.

536 **Statistical Downscaling and Disaggregation.** Statistical downscaling methods aim to infer high-
537 resolution estimates from coarse data, using spatial interpolation or learning-based techniques. These
538 have been widely applied in climate and environmental sciences [16, 38], including recent work on
539 probabilistic downscaling using Gaussian processes [17]. However, such methods typically model
540 associational patterns and do not support interventional or counterfactual reasoning.

541 To address this, some recent approaches integrate causal reasoning into downscaling pipelines, e.g.,
542 by identifying causally relevant predictors [18, 19]. Yet, these still rely on observable variables
543 and assume access to relatively fine-grained information. In contrast, our approach estimates high-
544 resolution *causal effects* using only aggregated outcome data and high-resolution covariates.

545 The challenges of reasoning with aggregate data are well-documented. Classical results on the
546 ecological fallacy show that aggregated statistics may obscure or even invert causal relationships [20].
547 Earlier work on aggregation and disaggregation in optimization similarly highlights the complexity
548 introduced when latent heterogeneity is present but unobserved [21]. CLAM explicitly addresses
549 these challenges by modeling the aggregation process and learning disentangled causal effects at the
550 subregional level.

551 **Causal (De)abstraction.** Structurally, our approach is inspired by recent work on causal abstraction,
552 which studies mappings between causal models at different levels of granularity [9, 24, 12]. These
553 works formalize when such mappings preserve counterfactual or interventional semantics, but do not
554 address learning disaggregated effects from data. Finally, our work relates to spatiotemporal causal

555 discovery [39], but differs by operating in a setting where temporal data is limited and only aggregate
556 observations are available.

557 **Graph Neural Networks.** Conceptually, our architecture is related to message-passing graph neural
558 networks (GNNs) [11]. In a GNN, pairwise messages are computed and aggregated within a single
559 update step. Similarly, our model learns an update function $f_{\theta}(\cdot)$ that operates on ordered inputs,
560 and optionally an aggregation function $g_{\phi}(\cdot)$ that is permutation-invariant, akin to a Deep Sets model
561 [10]. As in GNNs, we share the parameters of the structural function across nodes or regions.

562 **Invariant Causal Mechanisms.** Our work also relates to the principle of *invariant causal mecha-*
563 *nisms*, which posits that the functional relationships between variables remain stable across different
564 environments or interventions [13, 14]. This idea has been used for causal discovery [32] and domain
565 generalisation [33], and is particularly relevant for spatial disaggregation where intervention effects
566 may vary locally but share stable dependencies on contextual covariates. CLAM incorporates this
567 principle by enforcing that the learned local causal mechanism $f_{\theta}(\cdot)$ is shared across all subregions
568 while allowing spatial heterogeneity to emerge through contextual variables.

569 **Hierarchical Bayesian Model.** The design also parallels hierarchical Bayesian models [34, 35],
570 where group-specific effects are drawn from a shared prior distribution and information is partially
571 pooled across groups. In CLAM, the local mechanism $f_{\theta}(\cdot)$ plays the role of the shared prior, and
572 contextual covariates capture structured variation across subregions, allowing data-sparse subregions
573 to benefit from patterns learned in others.

574 **Spatiotemporal Causality.** Spatiotemporal causal inference has received growing attention as
575 researchers seek to understand how interventions and their effects evolve across both space and time. A
576 recent work presents a causal framework for studying the interplay between conflict and forest loss in
577 Colombia, emphasizing the need to model spatial and temporal dependencies jointly [25]. Extending
578 this perspective, methods proposed in [26] estimate heterogeneous treatment effects in settings where
579 economic assistance moderates the impact of airstrikes on insurgent violence, while approaches
580 introduced in [27] accommodate arbitrary spill-over and carryover effects across space and time.
581 Network-based settings with temporal dynamics have also been explored. In [28] the authors study
582 bipartite causal inference with interference in time series data, and in [31] methods are developed
583 to analyse the causal effects of modified treatment policies under network interference. Recent
584 methodological innovations include GST-UNet [29], a deep learning architecture for spatiotemporal
585 causal inference under time-varying confounding, and consistency-guided aggregation for causal
586 discovery in multivariate spatiotemporal data introduced in [30]. Together, these works highlight the
587 diversity of approaches to spatiotemporal causality, spanning structural models, interference-aware
588 estimation, and machine learning-driven inference.

Details of Exp. 1: Political Campaigning

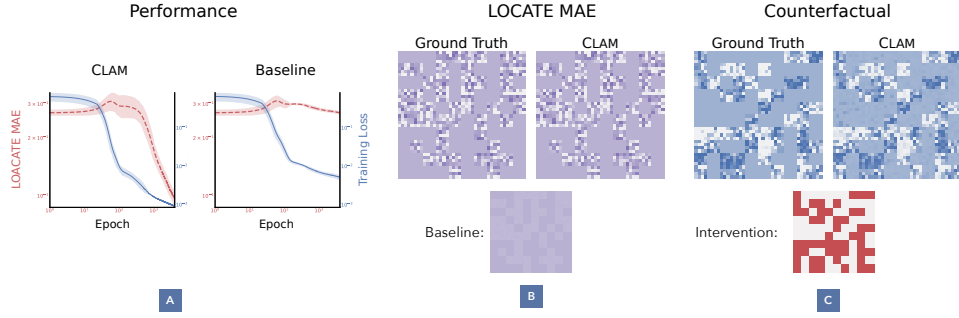


Figure 5: Evaluation of the learned outcomes, causal effect estimates, counterfactual predictions, and training behavior.

C Details of Main Experiments

C.1 Exp. 1: Political Campaigning

Context. We consider a simple political campaigning setting in which a candidate either spends campaign resources in a region or does not. The treatment is binary at the regional level and is inherited by all subregions. The outcome is the candidate’s relative improvement, but it is observed only as a regional aggregate. The effect of campaigning depends on subregional context, here interpreted as standardized wealth. Thus, two regions can have similar aggregate context and similar aggregate outcomes while still containing different subregional causal effects.

Setup. We use $N = 100$ regions arranged on a 10×10 grid. Each region contains $M = 16$ subregions arranged on a 4×4 grid. The treatment matrix is $T \in \{0, 1\}^{N \times M}$. Exactly half of the regions are treated, and within a treated region all subregions have treatment value 1; within a control region all subregions have treatment value 0.

The context matrix $C \in \mathbb{R}^{N \times M}$ contains one continuous contextual value per subregion. We sample the entries from a standard Gaussian distribution and then subtract the mean within each region, so that each row of C has mean zero. This makes the regional average context uninformative by construction. We then randomly swap entries across the matrix to introduce additional variation while preserving the overall structure. The observed outcome is generated at the subregional level and then aggregated to the region level by averaging.

The ground-truth outcome function is

$$f(t, c) = \begin{cases} 1, & t = 0, \\ \text{interp}(c), & t = 1, \end{cases}$$

where $\text{interp}(c)$ is the piecewise linear interpolation through the anchor points $\{(-2, 2), (0, 1), (1, 0)\}$.

Thus, untreated subregions have outcome 1, while treated subregions have an outcome that depends nonlinearly on the local context c . The noisy subregional outcome is

$$Y_{ij} = f(T_{ij}, C_{ij}) + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, 0.1^2),$$

and the observed regional outcome is

$$\hat{Y}_i = \frac{1}{M} \sum_{j=1}^M Y_{ij}.$$

The ground-truth local causal effect, which we use only for evaluation, is

$$E_{ij} = f(T_{ij}, C_{ij}) - f(0, C_{ij}).$$

Ablation Study of Exp. 1

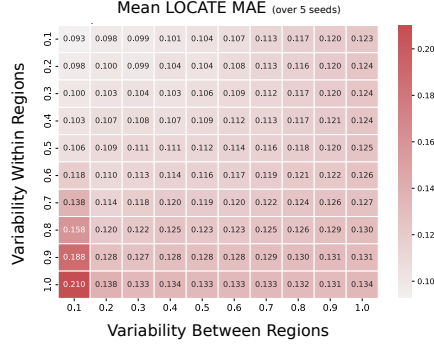


Figure 6: We evaluate the effect of varying the variance of regional context means and, for each fixed context mean, the inter-regional variance. Results are averaged over 5 random seeds for each parameter combination, and uncertainty bounds represent the corresponding variability.

Training. We train a small neural network $f_{\theta}(t, c)$ with two inputs, treatment and context, and one scalar output. The model has two hidden layers with 32 hidden units each, ReLU activations, dropout with probability 0.05, and is trained with AdamW using learning rate 10^{-3} and weight decay 10^{-3} for 4000 epochs. Importantly, CLAM is trained only from the regional aggregate signal. For each region, we average the model’s subregional predictions and minimize

$$\mathcal{L}_{\text{CLAM}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{M} \sum_{j=1}^M f_{\theta}(T_{ij}, C_{ij}) - \hat{Y}_i \right)^2.$$

Baseline. We compare against a uniform disaggregation baseline. Since the true subregional outcomes are not observed, this baseline assigns the same regional outcome to every subregion,

$$\tilde{Y}_{ij} = \hat{Y}_i.$$

It then trains the same neural network architecture using an element-wise loss,

$$\mathcal{L}_{\text{base}} = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M \left(f_{\theta}(T_{ij}, C_{ij}) - \tilde{Y}_{ij} \right)^2.$$

This gives the baseline a dense subregional loss signal, but the signal is based on a uniformity assumption that removes the within-region heterogeneity needed to recover local causal effects.

Evaluation. We evaluate whether the learned model can recover the fine-grained causal structure from aggregate observations (Figure 5). We report the training loss and LOCATE MAE (cf. Eq. (1)) over training (Figure 5a). Confidence intervals (95%) are bootstrapped and based on 20 runs. As expected, both decrease for CLAM, while the baseline cannot learn the causal mechanism. We also report the full LOCATE matrix, comparing the ground truth with the one inferred by CLAM, in Figure 5b. The recovered causal effect is almost indistinguishable from the ground truth, while the baseline, as expected, cannot reconstruct the underlying disaggregation. Lastly, we construct a counterfactual treatment matrix (Figure 5c) by inverting the observed treatment assignment, $T^{\text{cf}} = 1 - T$, and compare the ground-truth counterfactual outcome $f(T^{\text{cf}}, C)$ with the neural-network estimate $f_{\theta}(T^{\text{cf}}, C)$. This tests whether the learned mechanism supports coherent counterfactual prediction beyond the observed treatment assignment.

Ablation. To test when this recovery is possible, we repeat the experiment while varying two properties of the context matrix. The first is the standard deviation of the regional context means, denoted σ_{reg} . The second is the standard deviation of context values within each region, denoted σ_{subreg} . For each pair $(\sigma_{\text{reg}}, \sigma_{\text{subreg}})$, we run the experiment over multiple random seeds and average the final LOCATE MAE. This ablation shows that high variability within regions is associated with a smaller error.

Details of Exp. 2: Unknown Intervention Location

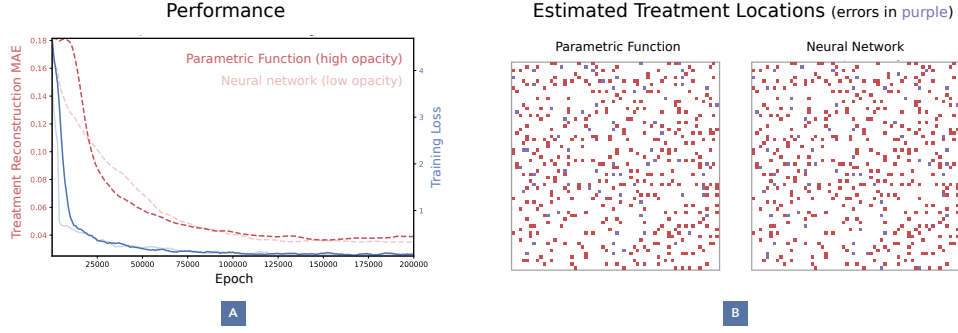


Figure 7: Performance of the training loss and MSE of the local-effect estimation, alongside ground-truth causal effects, estimated causal effects, and the corresponding estimation errors across subregions.

C.1.1 Ablation of Exp. 1: Political Campaigning

Our setting targets low-variability aggregate regimes, where regional outcomes are nearly constant and the inverse problem is underdetermined: many fine-grained causal explanations can yield indistinguishable regional observations. The key idea is that high-dimensional subregional covariates can restore information by exposing the shared causal mechanism $f_{\theta}(\cdot)$ to diverse contextual compositions within each aggregate unit. Exp. 1 and the corresponding ablation support this mechanism: as subregional heterogeneity increases, recovery of the true causal effect improves, even when aggregate outcomes remain similarly uninformative. By contrast, varying regional-level heterogeneity has a weaker effect, indicating that the gains do not come primarily from more variable aggregate outcomes, but from richer within-region contextual structure. See Figure 6 for details. Thus, causal deabstraction becomes possible in otherwise weakly identified settings when subregional covariate variation, together with an invariant causal mechanism, provides enough compositional diversity to distinguish the underlying fine-grained causal function.

C.2 Exp. 2: Public School Funding vs Improvement in Outcomes

Context. We examine how public spending on schools affects educational outcomes when the exact locations of the expenditures are *unknown*. Each administrative region contains multiple subregions, interpreted here as school districts, but only one district in a treated region receives a funding intervention. At the regional level, we only observe an aggregated treatment indicator and the aggregated outcome, not the specific district that received the intervention. The goal is to infer the high-resolution intervention locations from aggregated data.

The subregional context consists of high-resolution demographic and economic data, represented here by a normally distributed *wealth score*. We hypothesize that the effect of school funding depends on this wealth score through a global shift and scale.

Setup. We assume $N = 30 \times 30 = 900$ regions, each with $M = 2 \times 2 = 4$ subregions.

We simulate:

- A high-resolution intervention matrix $T \in \mathbb{R}^{N \times M}$, where $t_{i,j}$ indicates whether subregion j of region i receives the funding intervention.
- A regional intervention vector $\hat{T} \in \mathbb{R}^N$, where $\hat{T}_i = \sum_{j=1}^M t_{i,j}$ is the absolute intervention intensity in region i .
- A high-resolution context matrix $C \in \mathbb{R}^{N \times M}$ encoding the socioeconomic status of each subregion, where smaller $c_{i,j}$ values represent lower socioeconomic status and larger $c_{i,j}$ values represent higher socioeconomic status.
- A high-resolution noise matrix $\epsilon \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 = 0.005$.

675 The intervention matrix T is generated by:

- 676 1. Initializing all entries to zero.
- 677 2. Randomly selecting 50% of the regions for treatment.
- 678 3. For each treated region i , choosing exactly one subregion $j \in \{1, \dots, M\}$ uniformly at
- 679 random and assigning $t_{i,j} = 1$.

680 The context matrix C is sampled i.i.d. from a standard normal distribution:

$$c_{i,j} \sim \mathcal{N}(0, 1).$$

681 The outcome variable is given by the ground-truth functional relationship

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j} = (\theta_{\text{shift}} - c_{i,j}) \cdot \theta_{\text{scale}} \cdot t_{i,j} + \epsilon_{i,j},$$

682 with ground-truth parameters

$$\theta_{\text{shift}} = 4.2, \quad \theta_{\text{scale}} = 1.23.$$

683 The observed regional outcome is aggregated by summation:

$$\hat{Y}_i = \sum_{j=1}^M y_{i,j}.$$

684 As in Exp. 1, we assume no hidden confounding, no time-series effects, treatment assignment
685 independent of context, and aggregation-based regional observations.

686 **Training.** We train a model to estimate both the latent subregional intervention assignments and
687 the parameters of the causal effect function.

688 The model has:

- 689 • $N \times M = 900 \times 4$ latent treatment logits, one vector of length M for each region.
- 690 • Two scalar parameters $\{\theta_{\text{shift}}, \theta_{\text{scale}}\}$ specifying the parametric form of $f_{\theta}(\cdot)$.

691 The learned causal effect function is parametrized as

$$f_{\theta}(t_{i,j}, c_{i,j}) = (\theta_{\text{shift}} - c_{i,j}) \cdot \theta_{\text{scale}} \cdot t_{i,j}.$$

692 To ensure a positive scale parameter, θ_{scale} is represented through a softplus transformation.

693 An auxiliary preprocessing function is applied to the latent treatment logits using the Gumbel-
694 Softmax relaxation. This produces a soft row-wise assignment over the M subregions. The resulting
695 assignment is processed so that untreated rows remain all-zero and treated rows have approximately
696 one active subregion. During training, the Gumbel-Softmax temperature is exponentially annealed
697 from 2.0 to 0.1.

698 In the forward pass:

- 699 1. Sample a mini-batch of region indices.
- 700 2. Process the latent treatment logits using the Gumbel-Softmax relaxation to obtain an estimate
- 701 of the high-resolution treatment matrix T for the mini-batch.
- 702 3. Compute predicted subregional outcomes:

$$\hat{y}_{i,j} = f_{\theta}(\hat{t}_{i,j}, c_{i,j}).$$

- 703 4. Aggregate to regional predictions by summation:

$$\hat{Y}_i^{\text{pred}} = \sum_{j=1}^M \hat{y}_{i,j}.$$

- 704 5. Compare the regional prediction with the observed regional outcome $\hat{Y}_i$ using the MSE loss.

705 We optimize all parameters jointly using AdamW with learning rate 10^{-4} . Training is performed for
706 200,000 epochs with a fixed random seed.

Details of Exp. 3: Spatiotemporal Effects of Heat Waves on School Performance

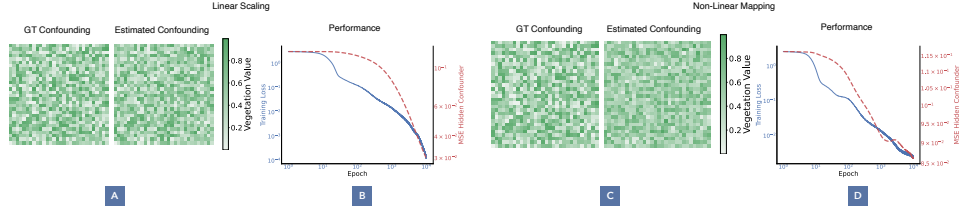


Figure 8: Ground-truth vegetation index, the corresponding estimated confounder fields, and joint plots of training loss and confounder MSE. Results are shown for two settings: linear scaling (left) and MLP learning (right).

Neural Network. Additionally, we train a neural-network variant following the same treatment-assignment procedure. In this version, the causal effect function $f_{\theta}(\cdot)$ is implemented as a feed-forward neural network taking $(t_{i,j}, c_{i,j})$ as input.

The network has three hidden layers with hidden dimension 16 and tanh activations:

$$(t_{i,j}, c_{i,j}) \mapsto \text{MLP}_{\theta}(t_{i,j}, c_{i,j}).$$

To stabilize the local-effect interpretation, the model uses the centered form

$$f_{\theta}(t_{i,j}, c_{i,j}) = \text{MLP}_{\theta}(t_{i,j}, c_{i,j}) - \text{MLP}_{\theta}(0, c_{i,j}),$$

which enforces $f_{\theta}(0, c_{i,j}) \approx 0$. The latent treatment indicators are processed in the same way as in the parametric model, using Gumbel-Softmax with temperature annealing from 2.0 to 0.1.

The neural-network model is also trained for 200,000 epochs using AdamW with learning rate 10^{-4} and mini-batches containing 20% of the regions.

Results. Figure 7a shows the evolution of the mini-batch training loss and the quality of the treatment-location reconstruction (measured as MAE). Both the neural network and the parametric function recover the locations reasonably well. We however find that this depends on the noise level and on proper temperature annealing. The final reconstructed locations are shown in Figure 7b, where incorrect predictions are shown in purple.

C.3 Exp. 3: Spatiotemporal Effects of Heat Waves on School Performance

Context. In this experiment, we examine the impact of extreme heat events (“heat waves”) on educational outcomes across both space and time. We model a spatial grid of $N = 100$ regions, each containing $M = 9$ subregions, observed over $W = 48$ consecutive months. The intervention represents the occurrence of a heat wave in a given subregion and month. Heat waves have a detrimental effect on school performance, but the strength of this effect depends on the educational background of parents in the subregion and on an unobserved environmental factor, vegetation coverage, which mitigates the impact of heat.

The observed context for each subregion is a categorical indicator of parents’ education level, taking one of three values (*low* = 1, *medium* = 2, *high* = 3), which evolves slowly over time to reflect gradual demographic changes. The unobserved confounding context is a fixed vegetation index between 0 and 1 for each subregion, representing the proportion of green cover and its protective effect against heat. The intervention process is binary at the subregional level, with most regions remaining unaffected in a given month, but some experiencing heat waves simultaneously in all their subregions, corresponding to large-scale weather patterns.

The outcome, measured as a monthly change in school performance for each subregion, is driven by the interaction between the intervention, parents’ education level, and the vegetation coverage. Subregions with lower parental education are more strongly affected, while vegetation dampens the negative effect. The challenge in this experiment is to recover both the mapping from context, intervention, and unobserved confounder to outcomes, and the vegetation values themselves, given only aggregated regional-level outcomes over time.

Setup. We model $N = 100$ regions, each with $M = 9$ subregions, observed over $W = 48$ discrete time points (months).

For each month $w \in \{1, \dots, W\}$, we define:

- A high-resolution binary treatment matrix $T^{(w)} \in \{0, 1\}^{N \times M}$, where $t_{i,j}^{(w)} = 1$ indicates that subregion j of region i is experiencing a heat wave in month w . In our data generation, each row of $T^{(w)}$ is either all zeros (probability 0.7) or all ones (probability 0.3), corresponding to large-scale heat events affecting entire regions.
- An observed context matrix $C^{(w)} \in \{1, 2, 3\}^{N \times M}$ encoding the categorical education level of parents in each subregion. At $w = 1$, these values are sampled uniformly. For $w > 1$, $C^{(w)}$ is obtained from $C^{(w-1)}$ by flipping each entry to a random category with probability 0.05, capturing slow demographic change.
- An unobserved covariate context matrix $U \in [0, 1]^{N \times M}$ containing fixed vegetation index values for each subregion, drawn uniformly from $[0, 1]$ and constant across all months.
- A noise matrix $\epsilon^{(w)} \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j}^{(w)} \sim \mathcal{N}(0, \sigma^2)$, with $\sigma^2 = 0.02$.

The subregional outcome is generated as:

$$y_{i,j}^{(w)} = f_{\theta}(t_{i,j}^{(w)}, c_{i,j}^{(w)}, u_{i,j}) + \epsilon_{i,j}^{(w)},$$

where $c_{i,j}^{(w)}$ is the parents' education category, $u_{i,j}$ is the vegetation index, and:

$$f_{\theta}(t_{i,j}^{(w)}, c_{i,j}^{(w)}, u_{i,j}) = \begin{cases} 0 & \text{if } t_{i,j}^{(w)} = 0, \\ (10 \cdot \mathbb{1}[c_{i,j}^{(w)} = 1] + 5 \cdot \mathbb{1}[c_{i,j}^{(w)} = 2] + \mathbb{1}[c_{i,j}^{(w)} = 3]) \cdot (1 - u_{i,j}) & \text{if } t_{i,j}^{(w)} = 1. \end{cases}$$

This function specifies that the subregional outcome depends on whether a heat wave occurs and, if so, how vulnerable the subregion is based on the education level of parents and the vegetation coverage. If $t_{i,j}^{(w)} = 0$, the effect is zero. If $t_{i,j}^{(w)} = 1$, the impact is determined by the categorical education level with coefficients (10, 5, 1) corresponding to low, medium, and high parental education, respectively, and is scaled by $(1 - u_{i,j})$, which reduces the effect in proportion to the amount of vegetation present. The indicator $\mathbb{1}[\cdot]$ selects the appropriate coefficient for each subregion.

The regional outcome at month w is obtained via mean aggregation. This setup induces a spatiotemporal causal problem with heterogeneous treatment effects driven by both observed and unobserved context, where the latter must be recovered from aggregated outcomes across multiple time points.

Training. We train a model to jointly estimate:

1. The mapping $f_{\theta}(\cdot)$ from observed context, intervention, and vegetation index to subregional outcomes,
2. The static vegetation index values $U \in [0, 1]^{N \times M}$ for all subregions.

The function $f_{\theta}(\cdot)$ is implemented as a 4 layer multilayer perceptron (MLP) with hidden dimension 16 and a dropout rate of 0.1 after the second hidden layer. Its inputs for each subregion are:

$$z_{i,j}^{(w)} = (\mathbb{1}[c_{i,j}^{(w)} = 1], \mathbb{1}[c_{i,j}^{(w)} = 2], \mathbb{1}[c_{i,j}^{(w)} = 3], t_{i,j}^{(w)}, u_{i,j}),$$

where $u_{i,j}$ is the vegetation index. The output is the predicted outcome $y_{i,j}^{(w)}$ for that subregion and month.

In the forward pass for a given month w :

1. The model takes as input $T^{(w)}$, $C^{(w)}$, and U ,
2. The MLP evaluates $f_{\theta}(\cdot)$ at each (i, j) to produce predicted subregional outcomes $y_{i,j}^{(w)}$,

- 778 3. These are aggregated to the regional level via mean aggregation.
- 779 4. The primary loss is the MSE between predicted and observed regional outcomes.

780 Training proceeds over all months in random order at each epoch. The total parameter set consists of
781 the MLP weights θ and the vegetation matrix U . We optimize using Adam with a learning rate of
782 0.001 for 10,000 epochs, with fixed random seeds for reproducibility. Loss curves for $\mathcal{L}_{\text{region}}$ and
783 the MSE of $\mathcal{L}_{\text{veg}}$ are recorded throughout training.

784 Results.

- 785 1. Figure 8a shows the loss curves and vegetation MSE under the linear scaling parameter-
786 ization. Both training loss and MSE decrease rapidly, and the final error is very small,
787 indicating near-perfect recovery of the unobserved covariate.
- 788 2. Figure 8b compares the ground-truth vegetation field to the estimated vegetation under linear
789 scaling. The two maps are almost indistinguishable, confirming that the restricted functional
790 form matches the data-generating process and allows highly accurate recovery.
- 791 3. Figures 8c and d present the same results for the non-linear MLP parameterization. While
792 the model still recovers meaningful structure in the vegetation, the MSE remains orders
793 of magnitude higher than in the linear case, and deviations from the ground truth are
794 clearly visible in the estimated field. This is notable since the true vegetation values are not
795 necessarily identifiable; the MLP could instead store a transformed version and map them
796 back during the forward process.
- 797 4. Overall, these results demonstrate that the model can reconstruct regional outcomes while
798 also uncovering hidden drivers of heterogeneity. The comparison between linear and non-
799 linear parameterizations highlights the trade-off: structural restrictions yield more accurate
800 recovery when they match the true mechanism, while flexible approximators such as MLPs
801 still learn the vegetation but with reduced precision.

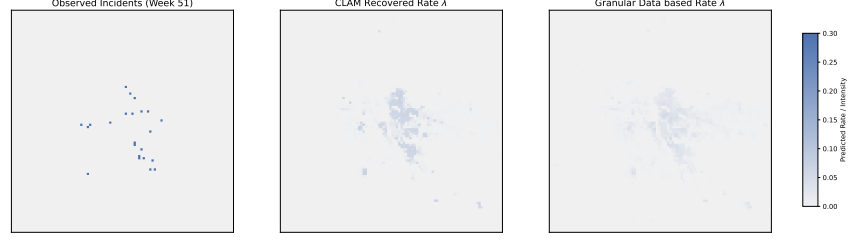


Figure 9: Comparison of learned spatial intensities for week 51. Both models are rendered on a shared color scale across the $\sqrt{N} \cdot M \times \sqrt{N} \cdot M$ grid, alongside the observed events for that week.

D Real-World Case Study Details: Effect of Heat on Gun Violence

This section provides the full details of the real-world study summarized in Section 6.2. The notation follows Section 2: regions are indexed by $i \in \{1, \dots, N\}$, subregions or cells by $j \in \{1, \dots, M\}$, and weeks by $w \in \{1, \dots, W\}$. Temperature is used as a continuous, time-varying treatment-like exposure; urbanization is the HR static contextual covariate; and gun-incident counts are the observed LR outcomes.

Problem Setting. The target variable is observed at a coarse spatial resolution, while covariates are available at a finer resolution. The study area is discretized into a fine grid of $\sqrt{N} \cdot M \times \sqrt{N} \cdot M$ cells, grouped into $\sqrt{N} \times \sqrt{N}$ coarse regions. With $N = 100$ and $M = 100$, this gives a 100×100 cell grid grouped into a 10×10 region grid. For each week w and region i , we observe a nonnegative count outcome

$$Y_{i,w}^{\text{LR}} \in \mathbb{Z}_{\geq 0}, \quad (2)$$

representing the number of gun-incident events occurring in region i during week w .

Data Representation and Covariates. Let $c_{i,j,w}$ denote the observed urbanization measurement for cell (i, j) and week w . Since urbanization is treated as time-invariant in this study, we use its temporal mean,

$$c_{i,j} = \frac{1}{W} \sum_{w=1}^W c_{i,j,w}. \quad (3)$$

Temperature is modeled at the region level. Let $t_{i,j,w}^{\text{HR}}$ denote the cell-level temperature for cell (i, j) and week w . We compute the region-average temperature exposure as

$$t_{i,w}^{\text{LR}} = \frac{1}{M} \sum_{j=1}^M t_{i,j,w}^{\text{HR}}, \quad (4)$$

and standardize it using the global mean and standard deviation across all region-week pairs:

$$\tilde{t}_{i,w} = \frac{t_{i,w}^{\text{LR}} - \mu_t}{\sigma_t + \varepsilon}, \quad (5)$$

where $\varepsilon > 0$ is a small constant for numerical stability. Missing temperature values are imputed using the global median temperature, and missing outcome counts are set to zero.

Disaggregation Model. We model the latent HR intensity, or rate, $\lambda_{i,j,w} > 0$ as a function of the standardized regional temperature exposure and the HR urbanization covariate:

$$\lambda_{i,j,w} = f_{\theta}(\tilde{t}_{i,w}, c_{i,j}), \quad (6)$$

where $f_{\theta}(\cdot)$ is a feed-forward neural network with parameters θ . To enforce nonnegativity, the network output is passed through a softplus transformation.

The model-implied LR expected count is obtained by summing the HR rates over all cells in region i :

$$\mu_{i,w} = \sum_{j=1}^M \lambda_{i,j,w}. \quad (7)$$

Thus, $\mu_{i,w}$ denotes the predicted regional count intensity, while $Y_{i,w}^{\text{LR}}$ denotes the observed regional count. This additive aggregation is consistent with the SCM in Section 3.1.

Training Objective. We assume a Poisson observation model at the region–week level,

$$Y_{i,w}^{\text{LR}} \sim \text{Poisson}(\mu_{i,w}), \quad (8)$$

and train f_{θ} by minimizing the negative Poisson log-likelihood, up to additive constants:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{W}_{\text{train}}| \cdot N} \sum_{w \in \mathcal{W}_{\text{train}}} \sum_{i=1}^N (\mu_{i,w} - Y_{i,w}^{\text{LR}} \log(\mu_{i,w} + \varepsilon)), \quad (9)$$

where $\mathcal{W}_{\text{train}}$ is the set of training weeks and $\varepsilon > 0$ is a small constant for numerical stability.

Spatial Intensity Comparison. Figure 9 visualizes observed incidents for a representative week, week 51, alongside spatial intensity estimates learned under different supervision regimes. When trained only on region-level counts, CLAM produces spatially structured intensity maps that concentrate mass around incident clusters but exhibit noticeably higher peak intensities and broader spatial support compared to the cell-supervised estimate. This overestimation reflects a fundamental ambiguity induced by aggregation: without access to fine-grained outcomes, the model must allocate regional mass across plausible subregional locations, leading to inflated local rates in areas consistent with the observed context and covariates. The response-surface comparison in the main paper, Figure 4, shows the same pattern in covariate space rather than physical space.

Learning. The training procedure optimizes the function parameters θ (or any other missing components of the SCM). The structural function $f_{\theta}(\cdot)$ can be represented by a neural network or any other parameterized function. Since the fine-grained outcomes $y_{i,j}$ are unobserved, training relies solely on their aggregated counterparts.

We optimize with the goal that the inferred aggregated predictions $(\hat{y}_i)$ match the observed aggregate: $\mathcal{L}_{\theta,\phi} = \sum_{i=1}^N (\hat{Y}_i - \hat{y}_i)^2$, where y_i denotes the ground truth of region i . If the noise terms are not i.i.d. and normally distributed, they can also be explicitly learned within the optimization loop.

Details of Exp. 4: Aggregation Function Estimation

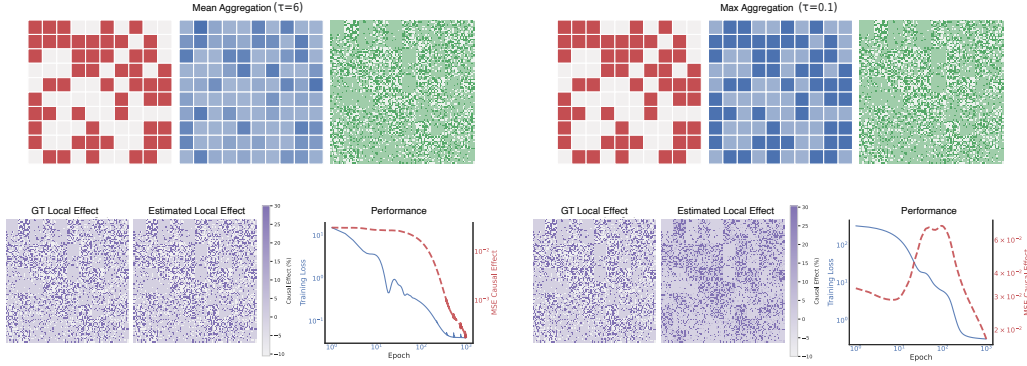


Figure 10: **Results Overview.** **Top:** Input of LR intervention locations (red), LR outcomes (blue), and HR context (green) for mean aggregation and max aggregation cases. **Bottom:** We observe the estimated local effects and the performance of our model over multiple epochs for both cases.

E Additional Experiments

Exp. 4 considers driving bans with an unknown aggregation function between subregional and regional outcomes. Exp. 5 introduces treatment–context confounding, where the probability of subregional treatment depends on contextual variables, affecting both allocation and outcomes

E.1 Exp. 4: Unknown Aggregation Functions

Context. We study the causal effect of driving bans on air quality under the assumption that the reported region-level air quality values are an unknown aggregation of finer-scale (subregional) measurements. Specifically, we model the aggregation as a learned combination of the subregional mean and maximum values, with the combination weights parameterized via a logit transform.

A realistic example is a city-wide driving ban where air quality is officially reported as a single regional index (e.g., $\text{PM}_{2.5}$ concentration), while the underlying measurement network collects data from multiple monitoring stations. Depending on reporting policies, the published index might resemble an average across stations, a worst-case station reading, or something in between.

The high-resolution context for each subregion is a single continuous variable: a vegetation index, capturing the amount of green coverage in the area. Vegetation can mitigate pollution and thus may modulate the effect of a driving ban. We hypothesize that the causal effect of driving bans varies with vegetation and that the unknown aggregation mechanism must be learned alongside the effect parameters to estimate subregional impacts correctly.

Setup. We assume $N = 100$ regions, each with $M = 100$ subregions.

We are given:

- A binary treatment vector $T \in \{0, 1\}^N$, where $T_i = 1$ indicates that a driving ban was implemented in region i (and thus all its subregions).
- A high-resolution context matrix $C \in [0, 1]^{N \times M}$ encoding the vegetation index of each subregion, where $c_{i,j} = 0$ represents minimal vegetation cover and $c_{i,j} = 1$ represents dense vegetation.
- A high-resolution noise matrix $\epsilon \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 = 0.02$.

The treatment vector T is generated using a Bernoulli distribution with probability 0.5, assigning each region to either the intervention or control group at random. The vegetation index values in C are drawn i.i.d. from $\text{Uniform}(0, 1)$.

The subregional outcome variable is given by the ground-truth functional relationship:

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j},$$

879 where:

- 880 • $t_{i,j} = T_i$ is the treatment status of subregion j in region i (inherited from the region),
- 881 • $c_{i,j}$ is the vegetation index,
- 882 • $\epsilon_{i,j}$ is Gaussian noise.

883 The causal effect function is parameterized as:

$$f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0.0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{veg}} \cdot c_{i,j} & \text{if } t_{i,j} = 1, \end{cases}$$

884 where θ_{base} captures the baseline effect of a driving ban and θ_{veg} captures how this effect changes
885 with vegetation cover.

886 Unlike Exp. 1, the regional outcome x_i is not a simple mean over subregions. Instead, we model the
887 aggregation function as:

$$y_i = \sum_{j=1}^M p_{i,j}(\tau) \cdot y_{i,j},$$

888 where:

$$p_{i,j}(\tau) = \frac{\exp(y_{i,j}/\tau)}{\sum_{k=1}^M \exp(y_{i,k}/\tau)}$$

889 is the softmax weight assigned to subregion j with temperature parameter $\tau > 0$. When $\tau \rightarrow \infty$,
890 $p_{i,j}$ approaches a uniform distribution (mean aggregation); when $\tau \rightarrow 0$, $p_{i,j}$ concentrates on the
891 subregion with the highest outcome (max aggregation).

892 **Training.** We train a model to estimate:

- 893 1. The parameters $\theta = \{\theta_{\text{base}}, \theta_{\text{veg}}\}$ of the causal effect function $f_{\theta}(\cdot)$,
- 894 2. The aggregation temperature parameter $\tau > 0$ governing the softmax-based aggregation
895 from subregional to regional outcomes.

896 The learned function takes the form:

$$f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0.0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{veg}} \cdot c_{i,j} & \text{if } t_{i,j} = 1. \end{cases}$$

897 In the forward pass:

- 898 1. Compute predicted subregional outcomes: $y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j})$,
- 899 2. Compute subregion-level aggregation weights via the softmax: $p_{i,j}(\tau) = \frac{\exp(y_{i,j}/\tau)}{\sum_{k=1}^M \exp(y_{i,k}/\tau)}$,
- 900 3. Compute predicted regional outcomes as the expectation under $p_{i,j}(\tau)$: $y_i = \sum_{j=1}^M p_{i,j}(\tau) \cdot$
901 $y_{i,j}$,
- 902 4. Compute the loss as the mean squared error with the observed Y_i .

903 All parameters $\{\theta_{\text{base}}, \theta_{\text{veg}}, \tau\}$ are optimized jointly using Adam with a learning rate of 0.001 for
904 1000 epochs. We initialize τ to a moderate value (e.g., $\tau = 1.0$) and enforce $\tau > 0$ during training
905 by optimizing its logarithm.

906 **Results.**

- 907 1. Visualizations of the aggregated causal effects show that mean aggregation preserves more
908 variation across regions, whereas max aggregation suppresses smaller effects and produces
909 sharper contrasts.
- 910 2. Although the subregional causal effect loss is never used directly for training, it decreases
911 steadily over epochs, indicating that the model recovers fine scale effects from only re-
912 gional supervision. Training loss is substantially higher under max aggregation, as weaker
913 subregional signals are obscured.

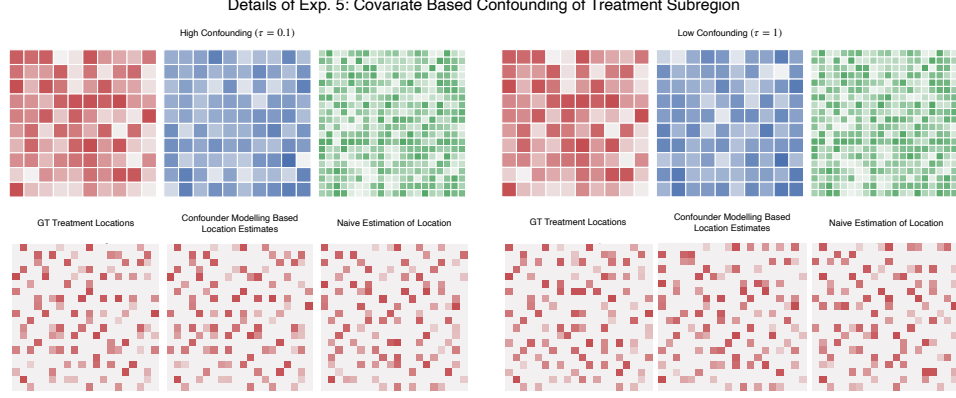


Figure 11: **Results Overview.** **Top:** Input of LR intervention locations (red), LR outcomes (blue), and HR context (green). **Bottom:** Ground truth treatment locations, estimation of treatment locations based on modeling of location covariate-based confounding, naïve estimation of treatment locations with all free parameters.

3. Comparing estimated and ground truth local effects confirms this bias: areas with small true effects tend to be overestimated when max aggregation dominates.
4. The aggregation function, parameterized by the temperature τ , is recovered with high accuracy. In the mean aggregation case, τ is estimated within 0.01 of the ground truth, while in the max aggregation case, the estimate is slightly underestimated (5.6), consistent with the plateauing behavior of the softmax function at low temperatures.
5. Overall, the experiment demonstrates that our approach can successfully recover both the aggregation mechanism (within its parametric form) and the underlying causal effects, despite only observing region-level outcomes.

E.2 Exp. 5: Covariate-Based Confounding of Treatment Subregion

Context. In this experiment, we study how confounding between treatment allocation and contextual variables affects causal effect estimation. We take the example of school districts receiving additional public funding and the corresponding changes in educational outcomes.

In reality, funding allocation is rarely random: wealthier or poorer districts may have systematically different chances of receiving funding due to political priorities, lobbying power, or policy constraints. This creates a dependency between the probability of treatment and district level context variables, violating the standard assumption of independent treatment assignment.

We model this confounding explicitly by making the probability of treatment allocation a deterministic function of the contextual variable, passed through a logistic transform to obtain valid probabilities for each region. The treatment effect itself is also a function of the same contextual variable, representing heterogeneous impacts across different socioeconomic conditions.

Setup. We assume $N = 100$ regions, each with $M = 100$ subregions (school districts).

We are given:

- A high-resolution treatment matrix $T \in \{0, 1\}^{N \times M}$ with *exactly one* treated subregion per region, i.e., $\sum_{j=1}^M t_{i,j} = 1$ for all i .
- A context matrix $C \in [0, 1]^{N \times M}$ encoding a socioeconomic/need index for each subregion, where a higher $c_{i,j}$ indicates greater need.
- A noise matrix $\epsilon \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$ (default $\sigma^2 = 0.02$).

Confounded Treatment Allocation. For each region i , we form logits from the subregional context,

$$\ell_{i,j} = \alpha_0 + \alpha_1 c_{i,j},$$

and convert them to a categorical distribution over subregions via a row-wise softmax,

$$p_{i,j} = \frac{\exp(\ell_{i,j})}{\sum_{k=1}^M \exp(\ell_{i,k})}.$$

We then draw exactly one treated subregion $j^* \sim \text{Categorical}(p_{i,1:M})$ and set

$$t_{i,j} = \mathbb{1}\{j = j^*\}, \quad \sum_{j=1}^M t_{i,j} = 1.$$

The parameters (α_0, α_1) control the strength and direction of confounding between context and treatment assignment (default $\alpha_0 = 0$, $\alpha_1 > 0$ so higher-need subregions are more likely to be treated).

Outcome Model With Context-Dependent Effects. Subregional outcomes follow the ground-truth functional relationship

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j},$$

with the causal effect function

$$f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{soc}} c_{i,j} & \text{if } t_{i,j} = 1, \end{cases}$$

where $\theta = \{\theta_{\text{base}}, \theta_{\text{soc}}\}$ captures the baseline funding effect and its modulation by socioeconomic need.

Regional Aggregation. Region-level outcomes are the sum over subregions (as in Exp. 1).

This construction induces confounding because (i) treatment allocation depends on $c_{i,j}$ through the softmax logits, and (ii) treatment effects also vary with $c_{i,j}$ via $f_{\theta}(\cdot)$.

Training. We compare two training regimes for estimating the causal effect parameters $\theta = \{\theta_{\text{base}}, \theta_{\text{soc}}\}$ and the high resolution treatment assignments $T \in \mathbb{R}^{N \times M}$ in the presence of treatment context confounding.

Regime 1: Naïve Training Under Independence Assumption. This setup directly applies the training procedure of Exp. 2 (Section C) to the confounded data, it:

1. Treats the treatment allocation as if independent of the context,
2. Learns T as free parameters, constrained to be one hot per row via a differentiable argmax with temperature,
3. Learns θ by minimizing the MSE between observed and predicted regional outcomes.

This ignores the fact that the treatment assignment mechanism is context dependent.

Regime 2: Confounding Aware Training. Here, we explicitly model the context-dependent treatment allocation mechanism. For each region i , we predict logits

$$\ell_{i,j} = \beta_0 + \beta_1 c_{i,j},$$

where (β_0, β_1) are learned parameters, and convert them to a categorical distribution via a row-wise softmax with a learnable temperature $\tau_{\text{conf}} > 0$:

$$p_{i,j}(\tau_{\text{conf}}) = \frac{\exp(\ell_{i,j}/\tau_{\text{conf}})}{\sum_{k=1}^M \exp(\ell_{i,k}/\tau_{\text{conf}})}.$$

We then use $p_{i,j}(\tau_{\text{conf}})$ as the (differentiable) treatment assignment in the forward pass.

In both regimes, predicted subregional outcomes are given by:

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{soc}} \cdot c_{i,j} & \text{if } t_{i,j} = 1, \end{cases}$$

972 where in Regime 1, $t_{i,j}$ comes from the naïvely learned $\hat{T}$, and in Regime 2, $t_{i,j} = p_{i,j}(\tau_{\text{conf}})$ from
973 the confounding model.

974 Regional outcomes are summed and aggregated. and the training loss is the mean squared error, All
975 parameters are optimized jointly using Adam with a learning rate of 0.001 for 1000 epochs. Regime 1
976 learns $\{\hat{T}, \theta_{\text{base}}, \theta_{\text{soc}}\}$, while Regime 2 learns $\{\beta_0, \beta_1, \tau_{\text{conf}}, \theta_{\text{base}}, \theta_{\text{soc}}\}$.

977 Results.

- 978 1. We compare two cases of confounding in subregional treatment allocation, holding region-
979 level treatments and contextual factors fixed. The aggregated causal effects differ markedly
980 depending on the level of confounding, showing that the treatment context dependency
981 directly shapes the observed regional outcomes.
- 982 2. We visualize the ground truth intervention locations alongside the estimates obtained with
983 and without explicitly modeling confounding. Under strong confounding, the confounding-
984 aware approach recovers the true intervention locations with high fidelity, while the naïve
985 method from Exp. 2 retrieves only a small fraction correctly.
- 986 3. In the low confounding setting, both methods perform similarly, and location probabilities
987 are estimated with comparable accuracy. This confirms that adjusting for confounding does
988 not reduce performance when confounding is weak.
- 989 4. For causal effect estimation, the final MSE of subregional effects under low confounding was
990 2.04 (naïve) versus 1.60 (confounding aware). Under high confounding, the respective errors
991 were 0.74 versus 0.32. These results show that accounting for confounding improves both
992 intervention location recovery and the accuracy of estimated causal effects by conditioning
993 on the correct contextual information.

994 Overall, this experiment demonstrates that explicitly modeling treatment context confounding is
995 essential as it enables accurate recovery of intervention locations and yields more reliable causal
996 effect estimates, particularly when treatment assignment is strongly biased by contextual factors.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction clearly state that the contribution is a framework for causal spatial disaggregation, illustrated through simulation studies and a real-world case study. Limitations are discussed in Section 7.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
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